

Paul S. Scotti, Ph.D

scottibrain@gmail.com | www.paulscotti.com | Oakland, CA

EXPERIENCE & EDUCATION

Sophont Inc. (https://sophont.med) Co-Founder and Chief Technology Officer	Mar. 2025 – Present
Building open multimodal medical foundation models, evaluation suites, and collaborative research infrastructure for healthcare.	
Princeton Neuroscience Institute Visiting research scientist (PI: Dr. Ken Norman) Postdoctoral research associate	Nov. 2023 – Present Apr. 2022 – Nov. 2023
Leading projects in real-time fMRI-to-image reconstruction, brain foundation models, and open neuroimaging data infrastructure.	
Stability AI Head of Neuroimaging & AI	Nov. 2023 – Jan. 2025
Led neuroimaging and AI research resulting in first-author NeurIPS and ICML papers on fMRI-to-image reconstruction; fine-tuned Stable Diffusion XL for state-of-the-art unCLIP performance.	
The Ohio State University Vision and Cognitive Neuroscience Lab Cognitive Control Lab (PI: Dr. Julie Golomb Dr. Andy Leber) Ph.D. dissertation on “Computational Models to Observe Visual Memory Distortions and Reconstruct Content from the Brain”	Oct. 2017 – Apr. 2022

The George Washington University Attention and Cognition Lab Visual Cognition Lab (PI: Dr. Sarah Shomstein Dr. Steve Mitroff) Undergraduate researcher (distinguished/honors scholar, magna cum laude, 2017 commencement speaker)	Sep. 2014 – May 2017
---	----------------------

GRANTS, FELLOWSHIPS, & AWARDS

- Princeton Innovation Fund for New Industrial Collaborations (\$250,000)
- NSF Graduate Research Fellowship (\$102,000)
- OSU University Fellowship (\$26,316)
- Luther Rice Undergraduate Research Fellowship

PRESS

- Our MindEye paper discussed in NY Times
- FujiFilm collaboration: mental imagery reconstruction
- Cognitive Revolution Podcast on mind reading
- Established industrial partnership between Stability AI x Princeton University to support neuroAI
- Our work in US Senate hearing on AI and IP

PROJECTS (recent selection)

CortexMAE: Scaling Vision Transformers for Functional MRI with Flat Maps
ICML 2026 family of fMRI foundation models trained on 2,100+ hours of open data using flat-map representations.
Brainmarks: Open Evaluation for fMRI Foundation Models
Open benchmark suite with more than 30 tasks across seven datasets, spanning brain-state decoding and clinical prediction.
Scaling Neuro: Open fMRI Data Infrastructure
Open infrastructure for sharing and processing fMRI datasets, with raw DICOM synchronization and upload-triggered compute.

Real-time MindEye: Reconstructing Visual Perception from fMRI

Real-time fMRI-to-image reconstruction system that produces continuously updated visual predictions during scanning.

Medmarks: Open Medical LLM Evaluation

Comprehensive open-source benchmark suite evaluating medical capabilities across 30 tasks and dozens of language models.

OpenMidnight: Training a Pathology Foundation Model for \$1.6K

Open recipe, model, and technical report showing that competitive pathology representation learning can be reproduced on a modest budget.

Labless and nanopath: Collaborative Autoresearch

Public research system and fast pathology challenge connecting human and autonomous experiments to reproducible source, metrics, and validation records.

MindEye and MindEye2: fMRI-to-Image Reconstruction

NeurIPS 2023 spotlight and ICML 2024 work on contrastive brain decoding, diffusion priors, and shared-subject models that reduce required training data by 40x.

SKILLS

- Python, PyTorch, multimodal foundation models, self-supervised learning, vision transformers, diffusion models, and LLM evaluation
- Distributed training and research infrastructure
 - DDP, FSDP, DeepSpeed, Slurm, Docker, AWS, Azure, and large-scale object-storage datasets
- Benchmarking and reproducible ML systems
 - Dataset curation, evaluation harnesses, experiment tracking, and autonomous research workflows
- Computational neuroimaging and neural decoding
 - fMRI, EEG, BIDS/DICOM, fMRIPrep, FSL, AFNI, SPM, FreeSurfer, and real-time fMRI
- Web and product engineering
 - HTML, CSS, JavaScript/TypeScript, React, and Node.js
- Hierarchical Bayesian modeling and behavioral experiment design

MENTORSHIP

Seungwan (Kevin) Son, Stephenie Chen, Karit (Keith) Matanachai, Ashutosh Narang, Cesar Torrico, Mihir Tripathy, Atmadeep Banerjee, Stepan Shabalín, David Weisberg, Foyez Alauddin, Nathalie Verlinde, Anisha Babu, Molly McKinney

AD HOC REVIEWING

ICML; CVPR; Nature Neuroscience; NeuroImage; Communications Biology; Imaging Neuroscience; Scientific Reports; Psychonomic Bulletin & Review; Journal of Experimental Psychology: General; Journal of Experimental Psychology: Learning, Memory, and Cognition; Attention, Perception, & Psychophysics; Memory; Memory & Cognition; Journal of Open Source Education

OUTREACH / PROFESSIONAL DEVELOPMENT / TEACHING

- MedARC open research community 2023 – Present
Organizing open NeuroAI projects and mentoring an international volunteer research community.
- Scaling Neuro 2026 – Present
Creating open infrastructure for sharing, processing, and reusing fMRI datasets.
- Labless and nanopath 2026 – Present
Building collaborative autoresearch infrastructure and a reproducible pathology foundation-model challenge.
- fMRI Playground: Simple summaries & simulations of neuroimaging methods 2023
Interactive textbook on computational neuroimaging methods using Python examples with simulated data

- OnNeuro, Founder 2017 – 2022
Hosting/sharing open-access research talks in the fields of psychology and neuroscience
- Center for Cognitive and Behavioral Brain Imaging Student Org, Technical Director 2017 – 2022
Organizing interdisciplinary workshops and guest speaker presentations at Ohio State Univ.
- Center for Cognitive and Brain Sciences Undergraduate Summer Institute (CUSI) 2018/2019/2021
Lectured on lab organization, questionable research practices, open science, and pre-registration
- NeuroHackademy Summer 2019
Led a team of researchers to create EduCortex, an educational brain viewer
- Guest Lecturer (Ohio State University) Fall 2019
Introduction to Psychology (PSYCH 1001)
- Course Assistant (Ohio State University) Spring 2019
Sensation and Perception (PSYCH 3310)
- Cognitive Psychology Laboratory (PSYCH 4510) 2018 – 2019
Introduction to Social Psychology (PSYCH 3325) Autumn 2018

PUBLICATIONS

1. Iyer, R. S., Tu, J. C., Villanueva, C. K. T., et al., Scotti, P. S., & Norman, K. A. (2026). Real-time reconstruction of human visual perception from fMRI. arXiv. arxiv.org/abs/2607.22753
2. Griot, M., Scotti, P. S., & Abraham, T. M. (2026). Compress-Distill: Reasoning trace compression for efficient knowledge distillation. arXiv. arxiv.org/abs/2606.05988
3. Kneeland, R., Villanueva, C. K. T., Ojeda, J., Khanna, S., Xu, J., Scotti, P. S., & Naselaris, T. (2026). MIRAGE: Robust multi-modal architectures translate fMRI-to-image models from vision to mental imagery. PLOS Computational Biology, in press. arxiv.org/abs/2605.17198
4. Warner, B., Grandhi, R. S., Kieffer, M., et al., Scotti, P. S., & Abraham, T. M. (2026). Medmarks: A comprehensive open-source LLM benchmark suite for medical tasks. ICML FM4LS Workshop. arxiv.org/abs/2605.01417
5. Lane, C., Tripathy, M., Murali, L. K., et al., Abraham, T. M., & Scotti, P. S. (2026). Scaling vision transformers for functional MRI with flat maps. Proceedings of the 43rd International Conference on Machine Learning. arxiv.org/abs/2510.13768
6. Kneeland, R., Jiang, W., Nunes, U. B., Scotti, P. S., Delorme, A., & Xu, J. (2026). ENIGMA: EEG-to-image in 15 minutes using less than 1% of the parameters. arXiv. arxiv.org/abs/2602.10361
7. Kaplan, D. Z., Grandhi, R. S., Lane, C., Warner, B., Abraham, T. M., & Scotti, P. S. (2025). How to train a state-of-the-art pathology foundation model with \$1.6k. Sophont Technical Report SOPHONT-TR-2025-001. doi.org/10.5281/zenodo.20711012
8. Xu, J., Nunes, U. B., Jiang, W., Ryther, S., Pringle, J., Scotti, P. S., Delorme, A., & Kneeland, R. (2025). Alljoined-1.6M: A million-trial EEG-image dataset for evaluating affordable brain-computer interfaces. NeurIPS Datasets and Benchmarks. arxiv.org/abs/2508.18571
9. Scotti, P. S., & Tripathy, M. (2025). Insights from the Algonauts 2025 winners. arXiv. arxiv.org/abs/2508.10784
10. Villanueva, C. K. T., Tu, J. C., Tripathy, M., Lane, C., Iyer, R. S., & Scotti, P. S. (2025). Predicting brain responses to natural movies with multimodal LLMs. arXiv. arxiv.org/abs/2507.19956
11. Kneeland, R., Scotti, P. S., Ghislain, S., Breedlove, J. L., Kay, K., & Naselaris, T. (2025). NSD-Imagery: A benchmark dataset for extending fMRI vision decoding methods to mental imagery. Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition. Spotlight; Outstanding Reviewer recognition to P. S. Scotti (top 5%). arxiv.org/abs/2409.01536

12. Luppi, A., Achterberg, J., Schmidgall, S., et al., Scotti, P. S., & Gellersen, H. (2024). Trainees' perspectives and recommendations for catalyzing the next generation of NeuroAI researchers. *Nature Communications*. doi.org/10.1038/s41467-024-53375-2
13. Scotti, P. S., Tripathy, M., Torrico, C., et al., Norman, K. A., & Abraham, T. M. (2024). MindEye2: Shared-subject models enable fMRI-to-image with 1 hour of data. *Proceedings of the 41st International Conference on Machine Learning*. arxiv.org/abs/2403.11207
14. Scotti, P. S., Banerjee, A., Goode, J., et al., Norman, K. A., & Abraham, T. M. (2023). Reconstructing the Mind's Eye: fMRI-to-image with contrastive learning and diffusion priors. *Advances in Neural Information Processing Systems*. Spotlight. arxiv.org/abs/2305.18274
15. Babu, A., Scotti, P. S., & Golomb, J. D. (2023). The dominance of spatial information in object identity judgments: A persistent congruency bias even amidst conflicting statistical regularities. *Journal of Experimental Psychology: Human Perception and Performance*. doi.org/10.1037/xhp0001104
16. Wallace, G., Polcyn, S., Brooks, P. P., et al., Scotti, P. S., Norman, K. A. (2022). RT-Cloud: A cloud-based software framework to simplify and standardize real-time fMRI. *NeuroImage*. doi.org/10.1016/j.neuroimage.2022.119295
17. Scotti, P. S., Chen, J., & Golomb, J. D. (2022). An enhanced inverted encoding model for neural reconstructions. *bioRxiv*. doi.org/10.1101/2021.05.22.445245
18. Scotti, P. S., & Maxcey, A. M. (2022). Directed forgetting of pictures of everyday objects. *Journal of Vision*. doi.org/10.1167/jov.22.10.8
19. Maxcey, A. M., Mancuso, E., Scotti, P. S., Spinelli, E., & Woodman, G. F. (2022). How to induce the forgetting of pictures. In W. Bainbridge & T. Brady (Eds.), *Visual Memory*. Routledge. ISBN 9780367744878.
20. Scotti, P. S., Kulkarni, A., Mazor, M., Klapwijk, E., & Huth, A. G. (2021). Interactive 3D brain helps you learn how the brain is organized. *Frontiers for Young Minds*. doi.org/10.3389/frym.2021.575131
21. Scotti, P. S., & Maxcey, A. M. (2021). What do laboratory-forgetting paradigms tell us about use-inspired forgetting? *Cognitive Research: Principles and Implications*. doi.org/10.1186/s41235-021-00300-6
22. Chen, J., Scotti, P. S., Dowd, E. W., & Golomb, J. D. (2021). Neural representations of task-relevant and task-irrelevant features of attended objects. *bioRxiv*. doi.org/10.1101/2021.05.21.445168
23. Scotti, P. S., Hong, Y., Leber, A. B., & Golomb, J. D. (2021). Visual working memory items drift apart due to active, not passive, maintenance. *Journal of Experimental Psychology: General*. doi.org/10.1037/xge0000890
24. Scotti, P. S., Hong, Y., Golomb, J. D., & Leber, A. B. (2021). Statistical regularities as a reference point for memory distortions: Swap and shift errors. *Attention, Perception, & Psychophysics*. doi.org/10.3758/s13414-020-02236-3
25. Scotti, P. S., Kulkarni, A., Mazor, M., Klapwijk, E., Yarkoni, T., & Huth, A. G. (2020). EduCortex: Browser-based 3D brain visualization of fMRI meta-analysis maps. *Journal of Open Source Education*, 3(26), 75. doi.org/10.21105/jose.00075
26. Scotti, P. S., Janakiefski, L., & Maxcey, A. M. (2020). Recognition-induced forgetting of schematically related pictures. *Psychonomic Bulletin & Review*, 27, 357-365. doi.org/10.3758/s13423-019-01693-8
27. Scotti, P. S., Collegio, A., & Shomstein, S. (2019). Object-based attention is resilient to low-level (boundary) or high-level (semantic) disturbances, but not both. *PsyArXiv*. doi.org/10.31234/osf.io/yxqju
28. Collegio, A., Nah, J., Scotti, P. S., & Shomstein, S. (2019). Attention scales according to inferred real-world object size. *Nature Human Behaviour*, 3(1), 40-47. doi.org/10.1038/s41562-018-0485-2

TALK / POSTER PRESENTATIONS (talks/workshops marked with *)

1. *Scotti, P. S. (2026, August 2). Scaling foundation models for fMRI and fairly benchmarking them. Modeling and Understanding Human Brain Computation at Scale, CCN 2026 Satellite Event. New York, NY. Invited talk (scheduled).
2. Lane, C., Tripathy, M., Murali, L. K., et al., Abraham, T. M., & Scotti, P. S. (2026, July). Scaling vision transformers for functional MRI with flat maps. ICML. Seoul, South Korea.
3. Warner, B., Grandhi, R. S., Kieffer, M., et al., Scotti, P. S., & Abraham, T. M. (2026, July). Medmarks: A comprehensive open-source LLM benchmark suite for medical tasks. ICML FM4LS Workshop. Seoul, South Korea.
4. Kneeland, R., Jiang, W., Nunes, U. B., Scotti, P. S., Delorme, A., & Xu, J. (2025, December). ENIGMA: EEG-to-image in 15 minutes using less than 1% of the parameters. NeurIPS Foundation Models for Brain and Body Workshop. San Diego, CA.
5. Xu, J., Nunes, U. B., Jiang, W., et al., Scotti, P. S., Delorme, A., & Kneeland, R. (2025, December). Alljoined-1.6M: A million-trial EEG-image dataset for evaluating affordable brain-computer interfaces. NeurIPS Datasets and Benchmarks. San Diego, CA.
6. Kneeland, R., Scotti, P. S., Ghislain, S., Breedlove, J. L., Kay, K., & Naselaris, T. (2025). NSD-Imagery: A benchmark dataset for extending fMRI vision decoding methods to mental imagery. Proceedings of the IEEE/CVF conference on computer vision and pattern recognition (CVPR). Nashville, TN.
7. Scotti, P. S., Tripathy, M., Torrico, C., Kneeland, R., Chen, T., Narang, A., Santhirasegaran, C., Xu, J., Naselaris, T., Norman, K. A., & Abraham, T. M. (2024). MindEye2: Shared-Subject Models Enable fMRI-To-Image With 1 Hour of Data. ICML. Vienna, Austria.
8. Scotti, P. S., Tripathy, M., Torrico, C., Kneeland, R., Chen, T., Narang, A., Santhirasegaran, C., Xu, J., Naselaris, T., Norman, K. A., & Abraham, T. M. (2024). MindEye2: Shared-Subject Models Enable fMRI-To-Image With 1 Hour of Data. ICLR Workshop on Representational Alignment (Re-Align). Vienna, Austria.
9. Scotti, P. S., Banerjee, A., Goode, J., Shabalin, S., Nguyen, A., Cohen, E., Dempster, A. J., Verlinde, N., Yundler, E., Weisberg, D., Norman, K. A., & Abraham, T. M. (2023). Reconstructing the Mind's Eye: fMRI-to-Image with Contrastive Learning and Diffusion Priors. NeurIPS. New Orleans, LA.
10. Scotti, P. S., Hennings, A. C., Wallace, G., Polcyn, S., Brooks, P. P., Mennen, A., Zhao, K., Michelmann, S., Li, K., Turk-Browne, N. B., Cohen, J. D., Norman, K. A. (2023). Cloud-based Software Framework to Simplify and Standardize Real-time fMRI. BRAIN Initiative. Bethesda, MD.
11. *Scotti, P. S., Hennings, A. C., Norman, K. A.. Conducting RT-fMRI Studies with the Realtime fMRI Cloud Framework (RT-Cloud). Real-Time Functional Imaging and Neurofeedback Meeting. New Haven, CT.
12. Wallace, G., Scotti, P. S., Polcyn, S., Brooks, P. P., Mennen, A., Zhao, K., Michelmann, S., Li, K., Turk-Browne, N. B., Cohen, J. D., Norman, K. A. (2022). Cloud-based Software Framework to Simplify and Standardize Real-time fMRI. BRAIN Initiative. Virtual conference.
13. Scotti, P. S., Chen, J., & Golomb, J. D. (2022, May). An enhanced inverted encoding model for neural reconstructions of visual perception, attention, and memory. Vision Sciences Society. Virtual conference.
14. Scotti, P. S., Chen, J., & Golomb, J. D. (2021, June). An improved method for evaluating inverted encoding models. Visual Working Memory Symposium. Virtual conference.
15. Scotti, P. S., Chen, J., & Golomb, J. D. (2021, May). An improved method for evaluating inverted encoding models. Vision Sciences Society. Virtual conference.
16. Chen, J., Scotti, P. S., Dowd, E. W., & Golomb, J. D. (2021, May). Neural representations of task-relevant and task-irrelevant features of attended objects. Vision Sciences Society. Virtual conference.
17. Scotti, P. S., Chen, J., & Golomb, J. D. (2021, March). An improved method for evaluating inverted encoding models. Cognitive Neuroscience Society. Virtual conference.

18. Jones, C. M., Scotti, P. S., & Golomb, J. D. (2020, May). Feature-binding errors during saccadic remapping may affect perception of real-world objects. Vision Sciences Society. Virtual conference.
19. Scotti, P. S., Kulkarni, A., Mazor, M., Klapwijk, E., Yarkoni, T., Huth, A. G. (2019, December). EduCortex: browser-based 3D brain visualization of fMRI meta-analysis maps. Awarded best poster, Center for Cognitive and Behavioral Brain Imaging Annual Research Days, Columbus, OH.
20. *Scotti, P. S., Hong, Y., Leber, A., B., & Golomb, J. D. (2019, November). Competition between similar visual working memory items underlies repulsion effects. Object Perception, Attention, and Memory (OPAM), Montreal, Quebec.
21. Scotti, P. S., Janakiefski, L., & Maxcey, A. M. (2019, November). Recognition-Induced Forgetting Does Not Operate Over Superordinate Categories. Psychonomic Society, Montreal, Quebec.
22. Scotti, P. S., Hong, Y., Leber, A., B., & Golomb, J. D. (2019, October). Competition Between Similar Visual Working Memory Items Produces Repulsion Effects. Society for Neuroscience, Chicago, IL.
23. Scotti, P. S., Hong, Y., Golomb, J. D., Leber, A., B. (2019, May). Relational interactions between visual memory representations increase with maintenance duration. Vision Sciences Society, St. Pete Beach, FL.
24. Babu, A., Scotti, P. S., Golomb, J. D. (2019, May). The dominance of spatial information in location judgments: A persistent congruency bias even amidst conflicting statistical regularities. Vision Sciences Society, St. Pete Beach, FL.
25. Janakiefski, L., Smerdell, M., Scotti, P. S., Maxcey, A. (2019, March). Does recognition-induced forgetting operate over temporally-grouped objects? CogFest, Columbus, OH.
26. Scotti, P. S., Hong, Y., Golomb, J. D., Leber, A., B. (2018, November). Statistical regularities during object encoding distort long-term memory. Awarded best poster (\$200), Object Perception, Attention, and Memory (OPAM), New Orleans, LA.
27. Scotti, P. S., Hong, Y., Golomb, J. D., Leber, A., B. (2018, September). Statistical regularities during object encoding distort long-term memory. Center for Cognitive and Brain Sciences Fall Retreat, Mt. Sterling, OH.
28. Scotti, P. S., Hong, Y., Golomb, J. D., Leber, A., B. (2018, May). Statistical regularities during object encoding distort long-term memory. Vision Sciences Society, St. Pete Beach, FL.
29. Adamo, S., Nah, J., Collegio, A., Scotti, P. S., Shomstein, S. (2018, May). The flux capacitor account: A new theoretical account of multiple target visual search errors. Vision Sciences Society, St. Pete Beach, FL.
30. *Collegio, A., Nah, J., Scotti, P. S., Shomstein, S. (2017, November). Real-world object size affects attentional allocation. Object Perception, Attention, and Memory (OPAM), Vancouver, BC.
31. Scotti, P. S., Collegio, A., & Shomstein, S. (2017, November). Task-irrelevant object category guides attentional allocation. Object Perception, Attention, and Memory (OPAM), Vancouver, BC.
32. Scotti, P. S., Adamo, S., Mitroff, S., Shomstein, S. (2017, May). Repetition priming preferentially benefits infrequent targets. Vision Sciences Society, St. Pete Beach, FL.
33. Adamo, S., Nah, J., Collegio, A., Scotti, P. S., Shomstein, S. (2017, May). Does orientation matter? Same or differently oriented targets in a multiple target search. Vision Sciences Society, St. Pete Beach, FL.
34. Collegio, A., Nah, J., Scotti, P. S., Shomstein, S. (2017, May). Real-world object size affects attentional allocation. Vision Sciences Society, St. Pete Beach, FL.
35. Scotti, P. S., Adamo, S., Mitroff, S., Shomstein, S. (2017, April). Repetition priming preferentially benefits infrequent targets. 1st place Psychology poster, GW Research Days event, Washington, D.C.
36. Scotti, P. S., Malcolm, G.L., Peterson, M., & Shomstein, S. (2016, November). Reality vs. Simplicity: The effects of real-world objects on attentional selection. Object Perception, Attention, and Memory (OPAM), Boston, MA.
37. Scotti, P. S., Malcolm, G.L., Peterson, M., & Shomstein, S. (2016, May). Reality vs. Simplicity: The effects of real-world objects on attentional selection. Vision Sciences Society, St. Pete Beach, FL.